

صَرَح

SARH

المخططات الفنية للنظام

SYSTEM DIAGRAMS · CLASSIC UML · A4 EDITION

LVCT — Libya Vision for Communication & Technology
Version 1.1 · 2026-05-07

المحتويات · Contents

1. القسم 1 · Use case & org chart

2. 1.1 Use-case diagram — مخطط حالات الاستخدام

3. 1.2 Organisational chart — الهيكل التنظيمي

4. القسم 2 · Conceptual + logical ERD

5. 2.1 Conceptual ERD (Chen-style) — المخطط المفاهيمي

6. 2.2 Logical ERD · Identity cluster

7. 2.3 Logical ERD · Property cluster

8. 2.4 Logical ERD · System cluster

9. القسم 3 · Data dictionary (tree)

10. 3.1 Identity domain

11. 3.2 Property + On-chain domain

12. 3.3 System domain

13. القسم 4 · Class diagram & data flow

14. 4.1 Class diagram (.NET 8 backend)

15. 4.2 DFD · Level 0 (context)

16. 4.3 DFD · Level 1

17. 4.4 DFD · Level 2 (spatial)

18. 4.5 System flow (end-to-end)

19. القسم 5 · Sequence diagrams

20. 5.1 Sign-in

21. 5.2 Property approval (1/3) · Submit + Review + Manager approval

22. 5.3 Property approval (2/3) · Deed PDF + SSI + IPFS + NFT mint

23. 5.4 Property approval (3/3) · Delivery + public verify

24. 5.5 Property approval · full reference (overview)

25. 5.6 ID-card issuance (5-step wizard)

26. القسم 6 · User journeys

Citizen journey 6.1 .27

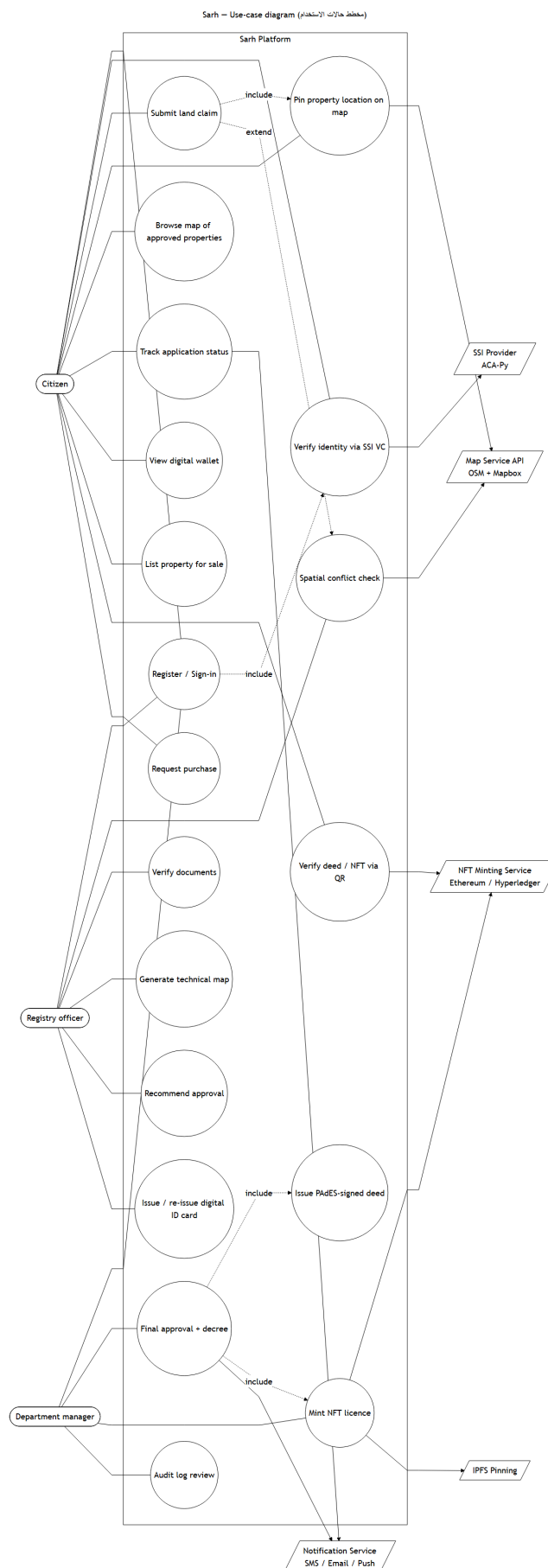
Officer + Manager journey 6.2 .28

01

حالات الاستخدام والهيكـل التنظيمي

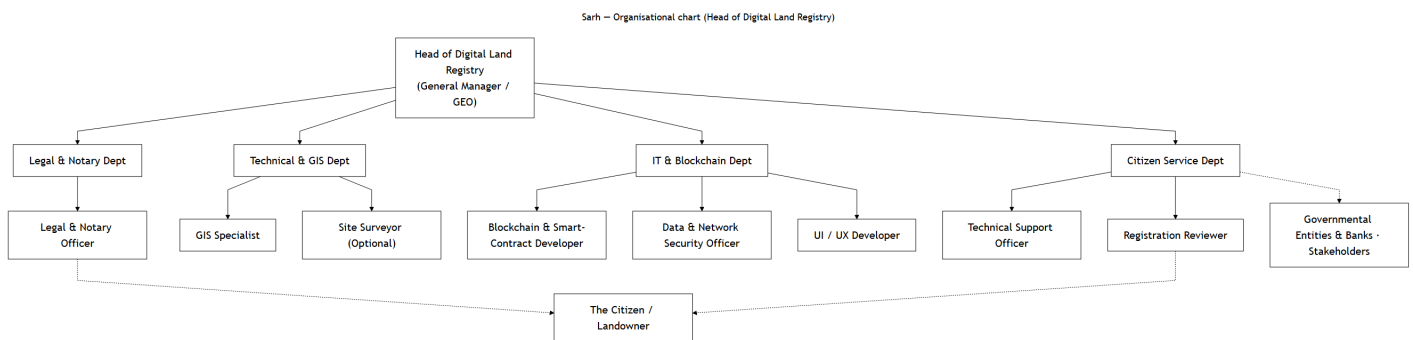
USE CASE & ORG CHART

Maps the actors and use-cases the platform exposes, plus the
.organisational structure of the Digital Land Registry



يوضّح المخطط الفاعلين الرئيسيين (المواطن، موظف السجل، مدير القسم) و١٩ حالة استخدام داخل المنصة، مع الأنظمة الخارجية (SSI، Map، Notification، NFT minting، IPFS).

Citizen / Registry officer / Department manager actors against 19 use-cases inside the platform, with external systems.



رئيس مصلحة السجل العقاري الرقمي ← أربعة أقسام (الشؤون القانونية والتوثيق، التقني وGIS، تقنية المعلومات والبلوكتشين، خدمة المواطن) ← الأدوار، مع المواطن وأصحاب المصلحة كأطراف خارجية.

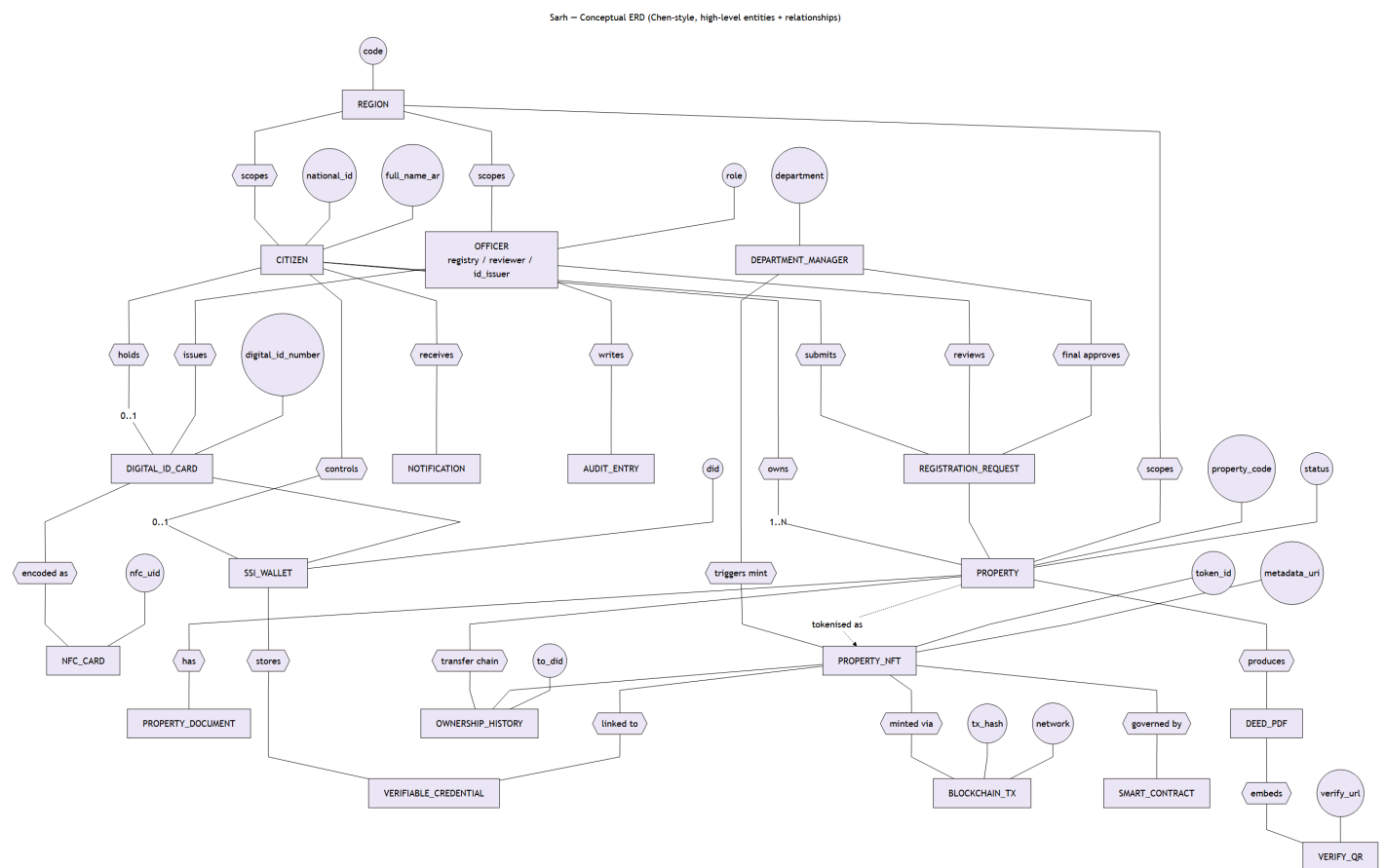
Head of Digital Land Registry → 4 departments → individual roles, with the Citizen / Landowner and Stakeholders shown as external actors.

02

المخطط المفاهيمي والمنطقي

CONCEPTUAL + LOGICAL ERD

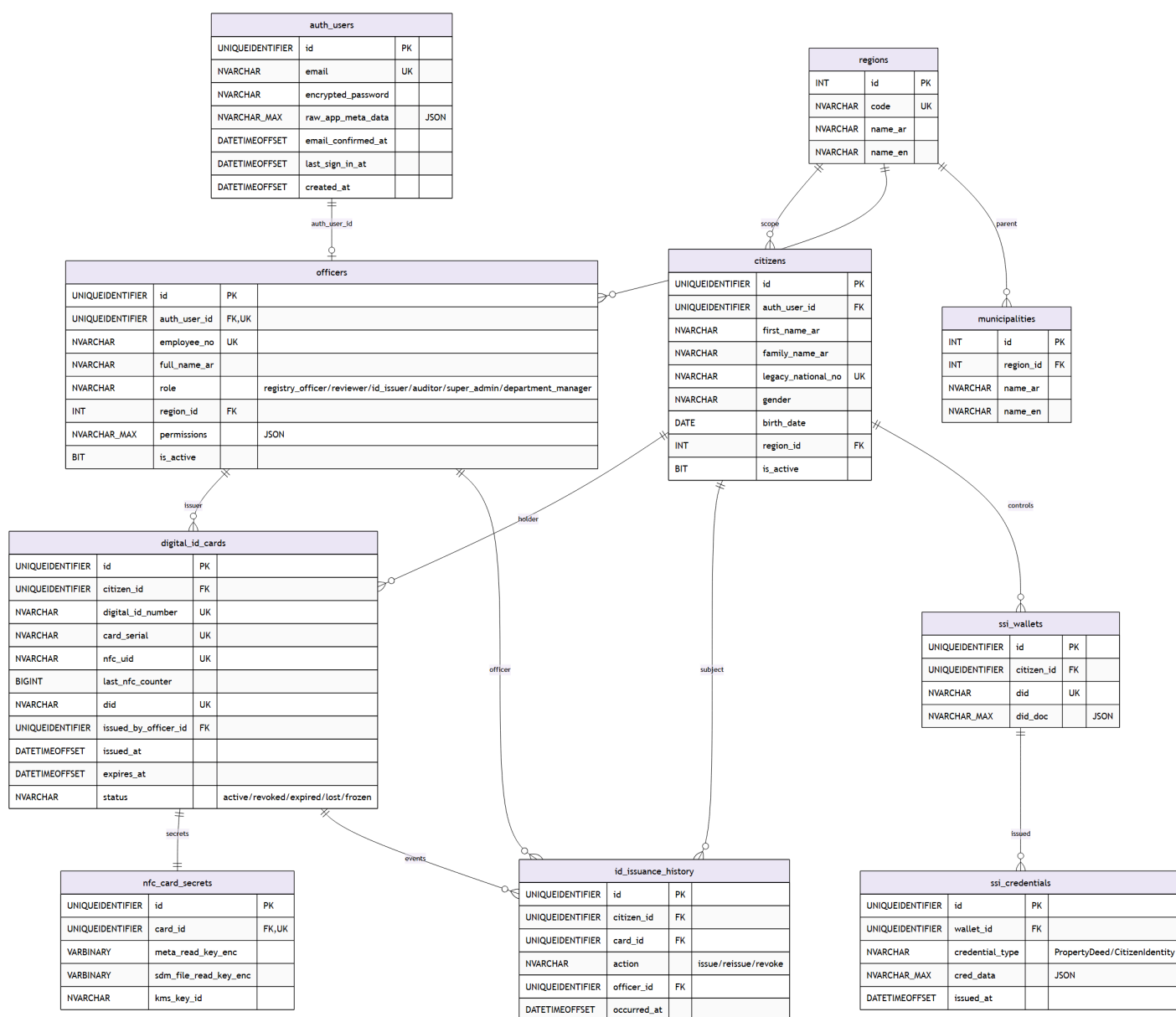
Chen-style entities + relationships, and the full SQL Server schema split into
.three printable clusters with explicit PK / FK / UK markers



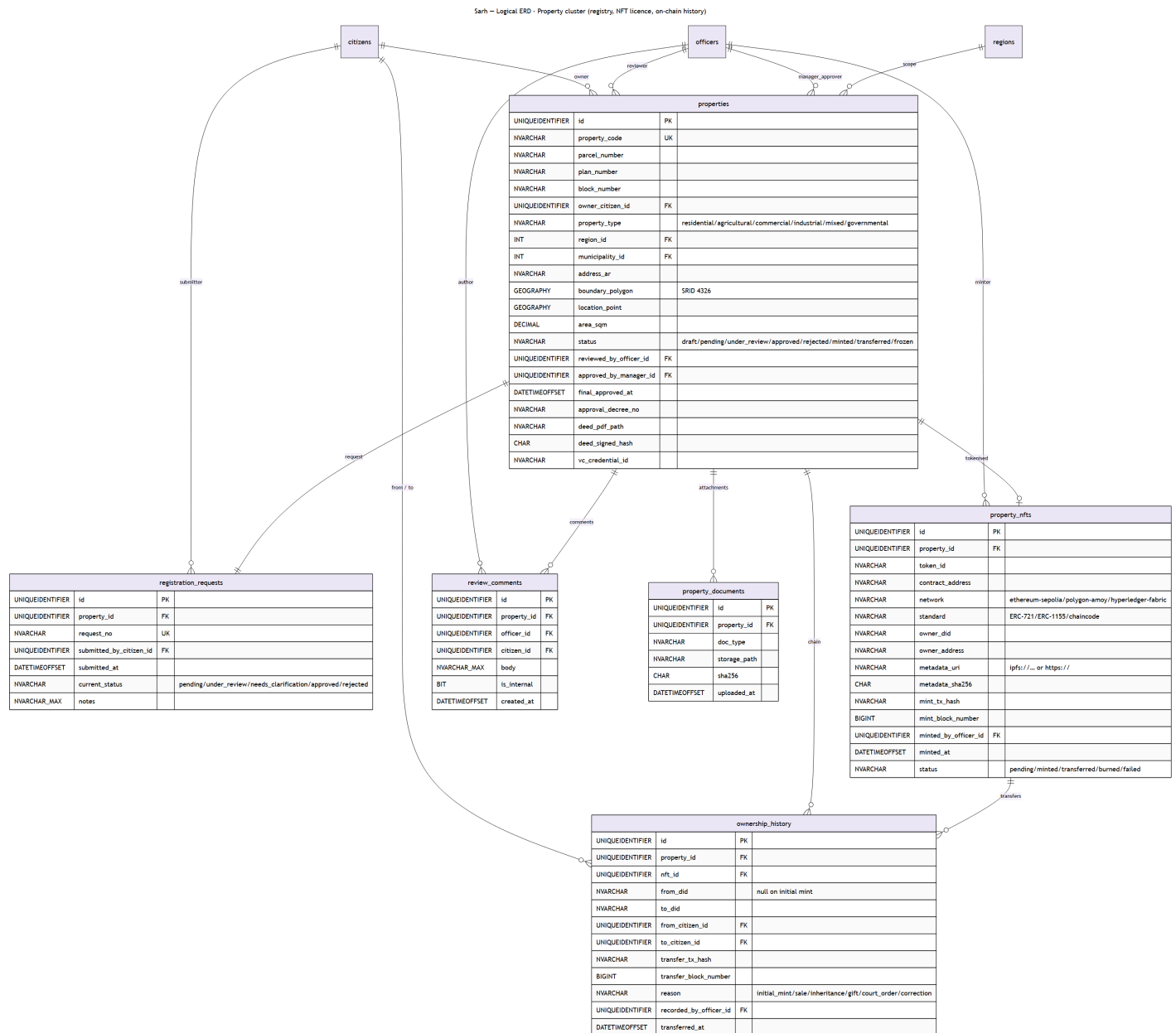
كيانات (مستطيلات)، علاقات (معينات)، وصفات أساسية (بيضاوي). يشمل PROPERTY_NFT و OWNERSHIP_HISTORY و SMART_CONTRACT و BLOCKCHAIN_TX و DEPARTMENT_MANAGER الجديد.

Chen notation — entities (rectangles), relationships (diamonds), key attributes (ovals). Includes the on-chain licence flow and the third human role.

Sarh — Logical ERD · Identity cluster (auth, citizens, officers, ID cards, SSI)

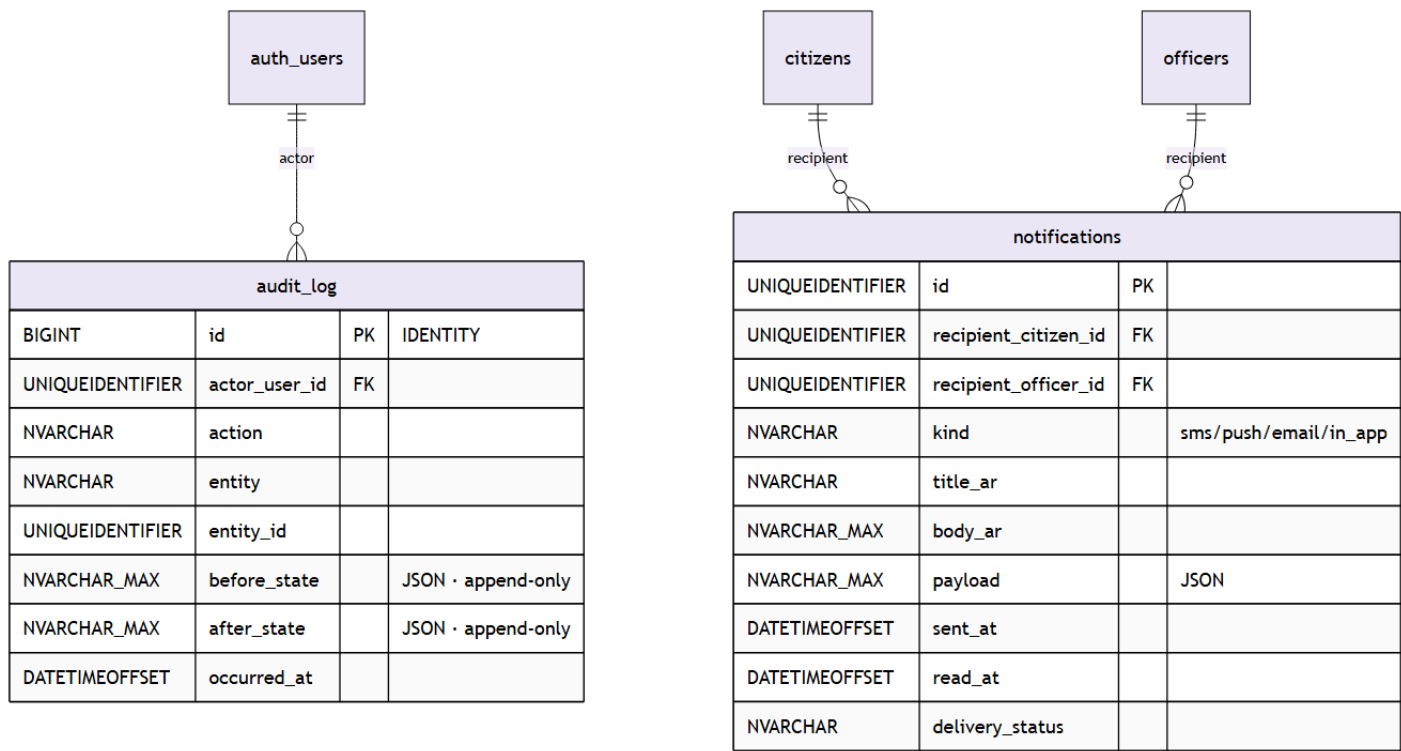


auth_users, citizens, officers, regions, municipalities, digital_id_cards, nfc_card_secrets, ssi_wallets, ssi_credentials, id_issuance_history.



properties, registration_requests, review_comments, property_documents, property_nfts, ownership_history.

Sarh — Logical ERD · System cluster (notifications, audit log)



notifications + audit_log (append-only via INSTEAD-OF triggers).

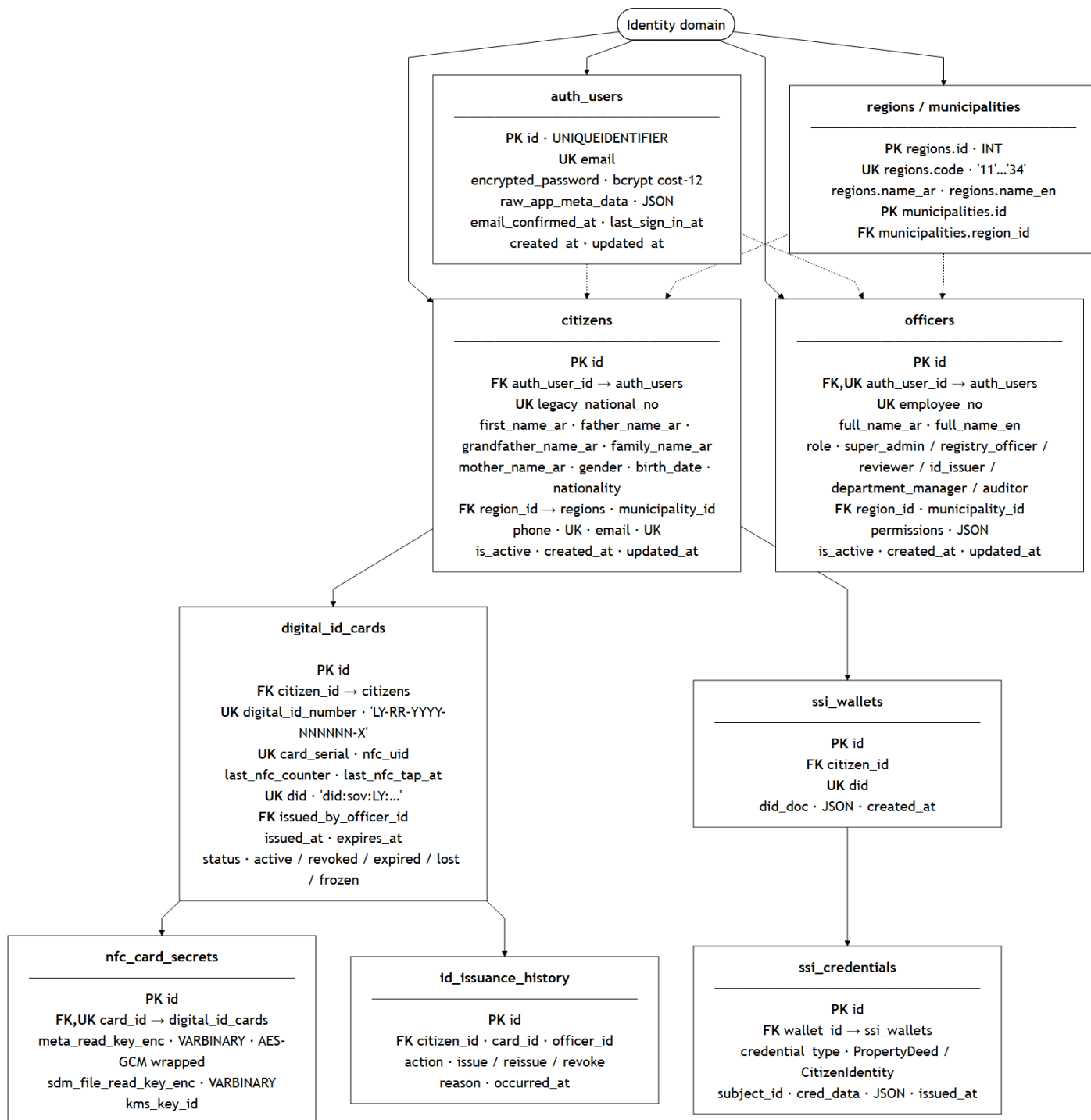
03

قاموس البيانات (شجرة)

DATA DICTIONARY · TREE VIEW

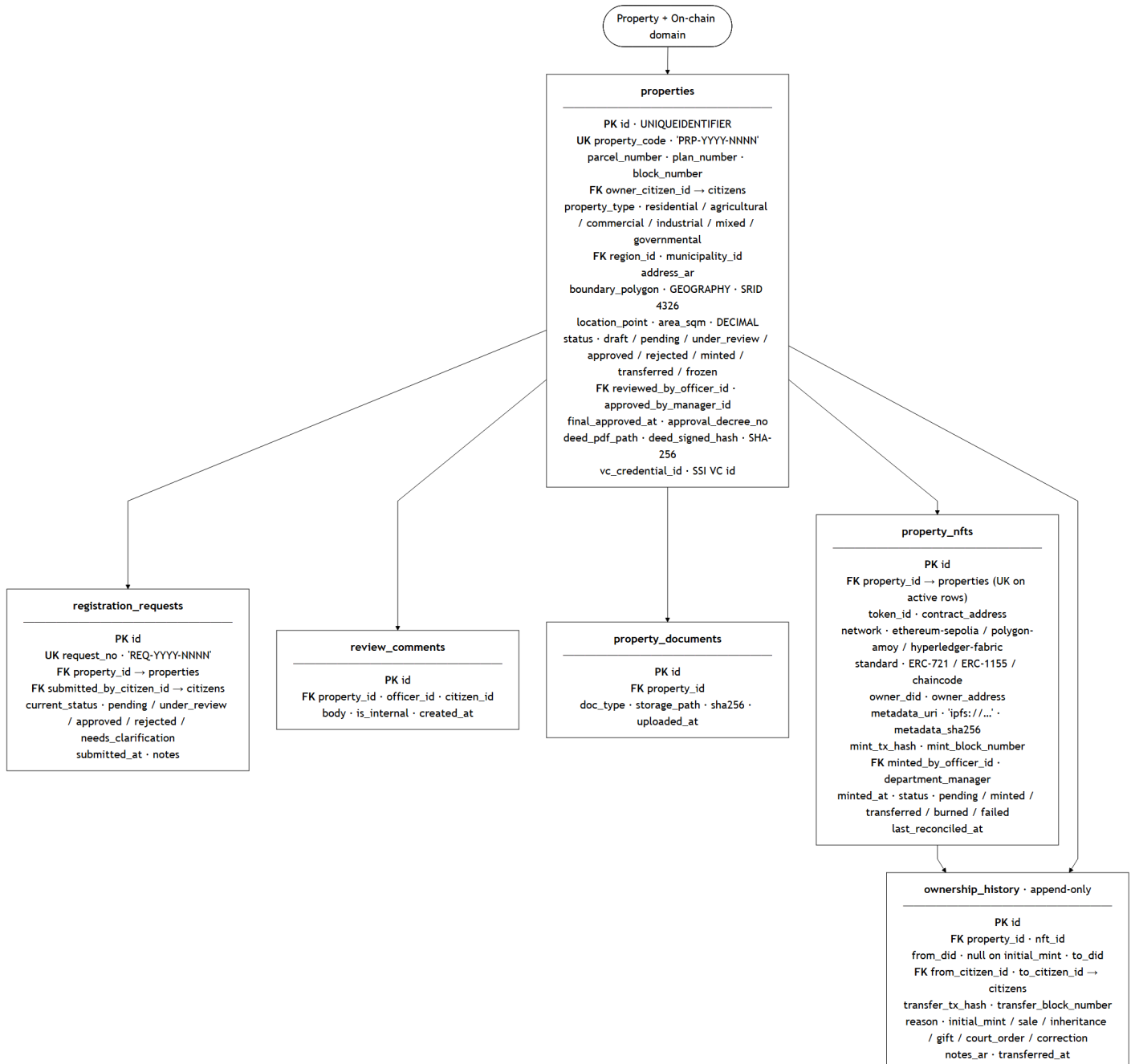
Each table rendered as one rich node showing PK / FK / UK markers, the data type, and the workflow enum values. Three printable trees split by .domain

Sarh — Data Dictionary · Identity domain



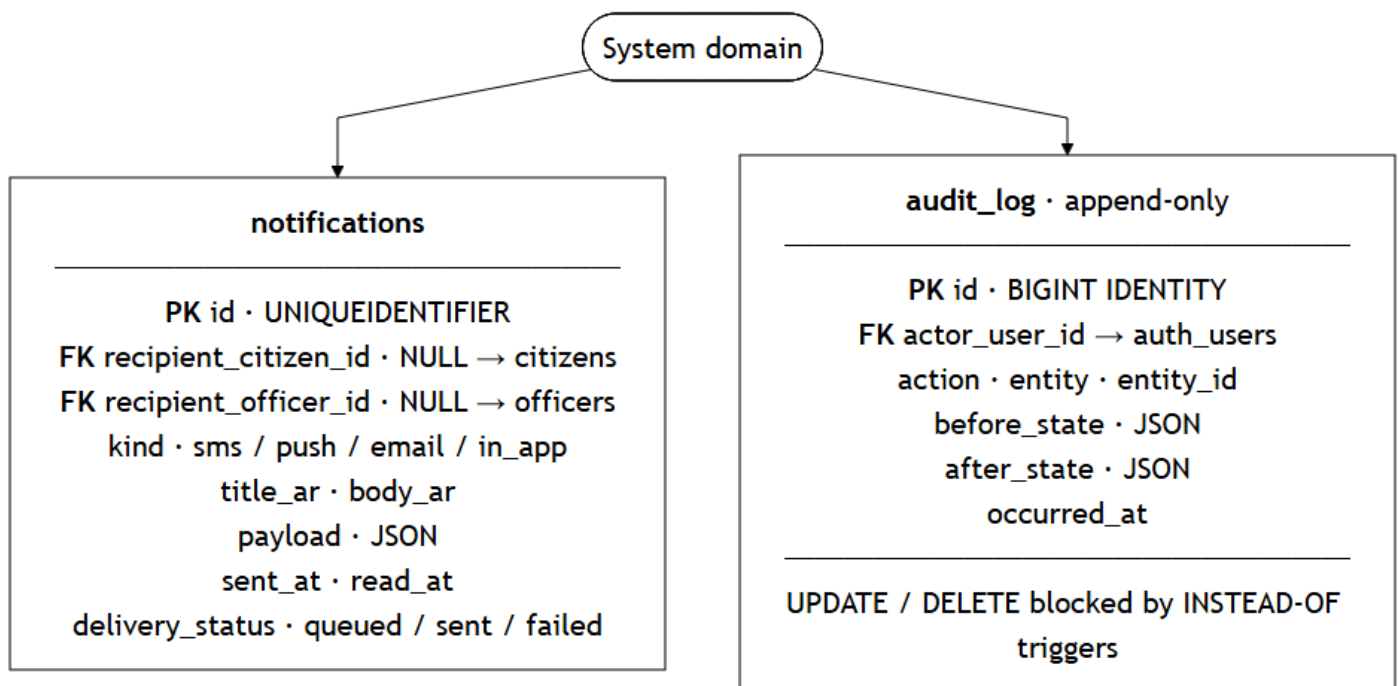
auth_users · citizens · officers · regions · digital_id_cards · ssi_*

Sarh — Data Dictionary · Property + On-chain



properties · registration_requests · review_comments · property_documents · property_nfts · ownership_history.

Sarh — Data Dictionary · System

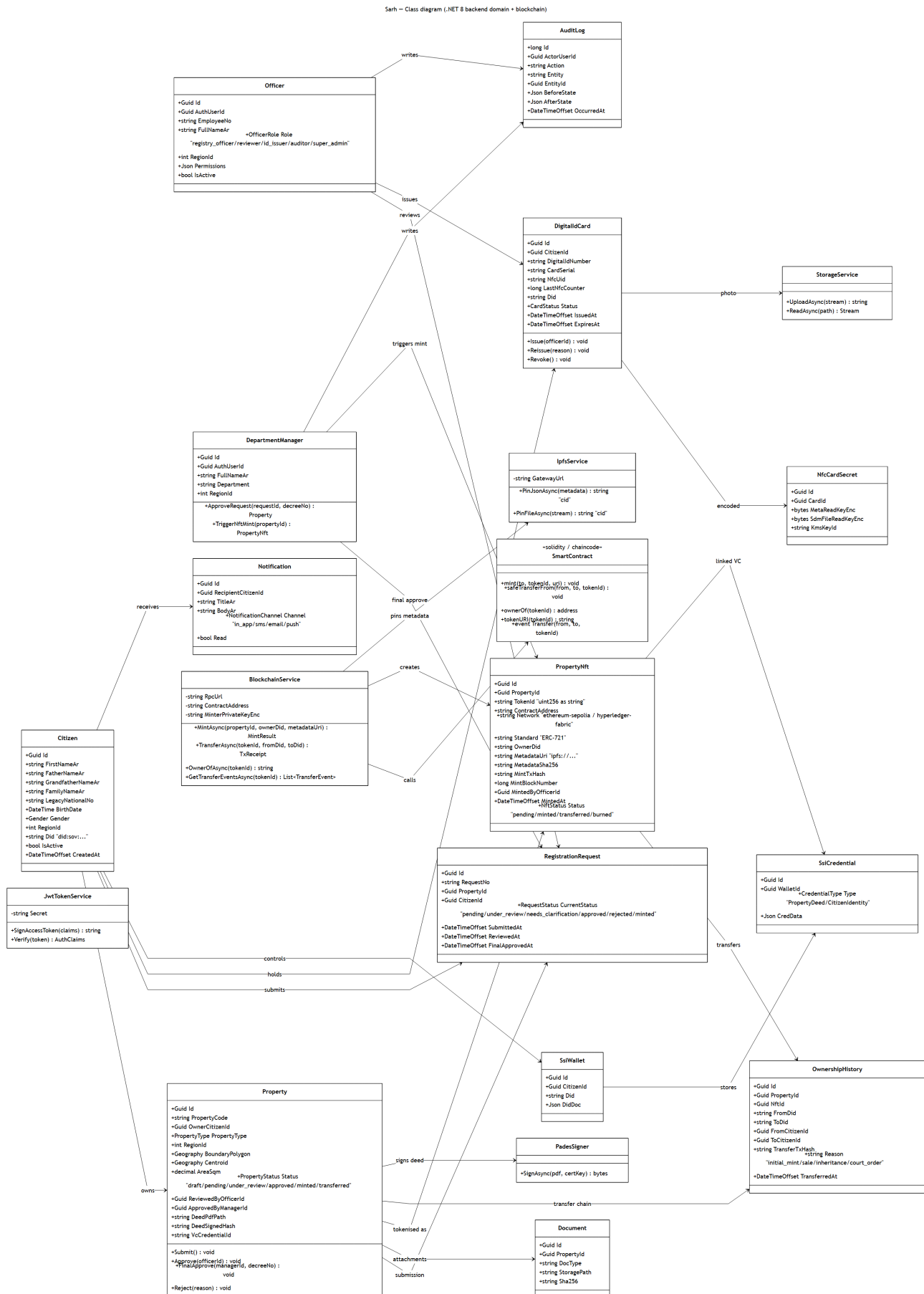


04

مخطط الفئات وتدفق البيانات

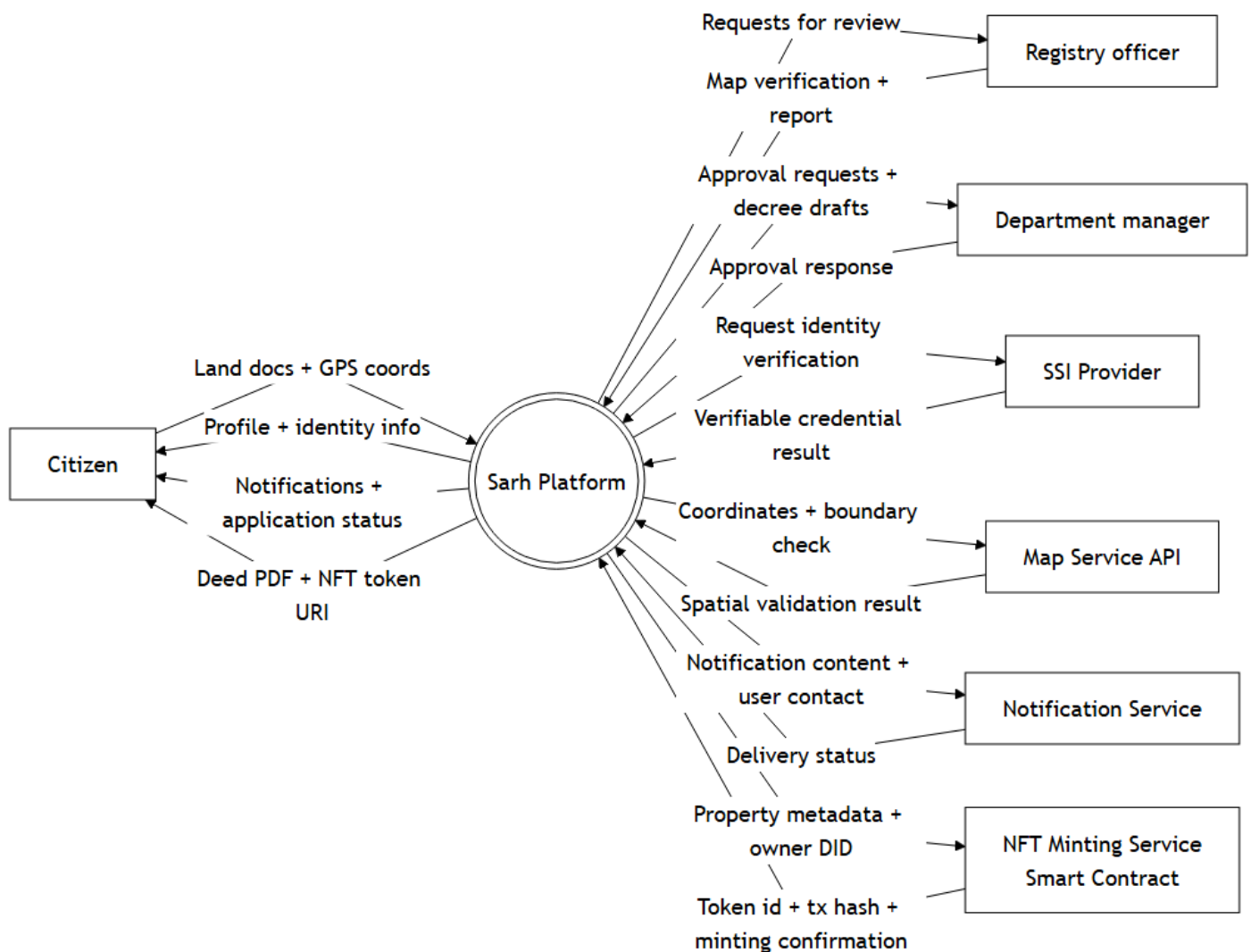
CLASS DIAGRAM & DFD

The .NET 8 backend domain expressed as a UML class diagram, and three levels of data-flow — context, sub-processes, and the spatial-audit .decomposition

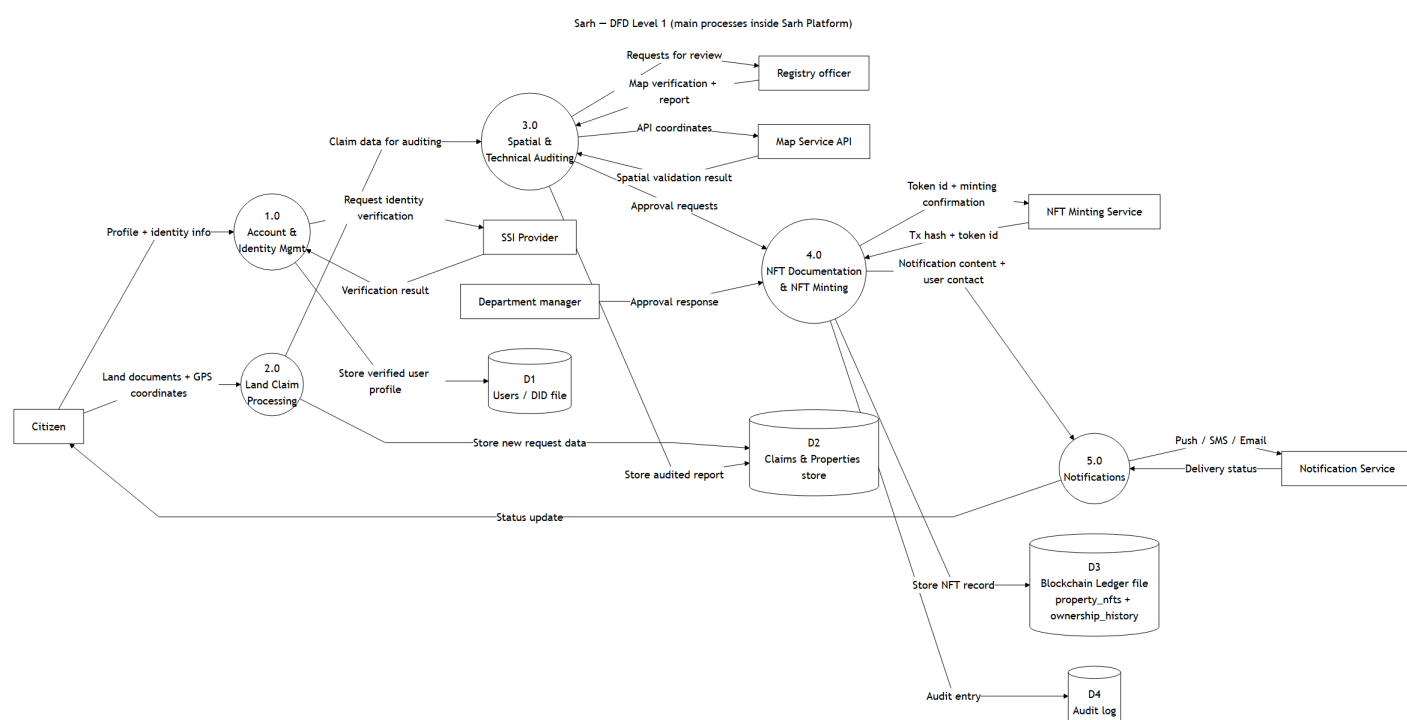


Citizen, Officer, DepartmentManager, Property, RegistrationRequest, PropertyNft, OwnershipHistory, BlockchainService, SmartContract, IpfsService, DigitalIdCard, NfcCardSecret, SsiWallet, SsiCredential, JwtTokenService, StorageService, PadesSigner.

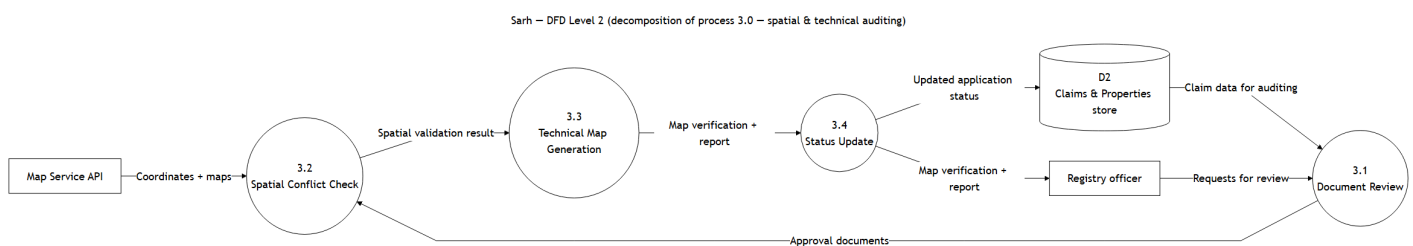
Sarh — Data-flow Context diagram (DFD Level 0 / Single process)



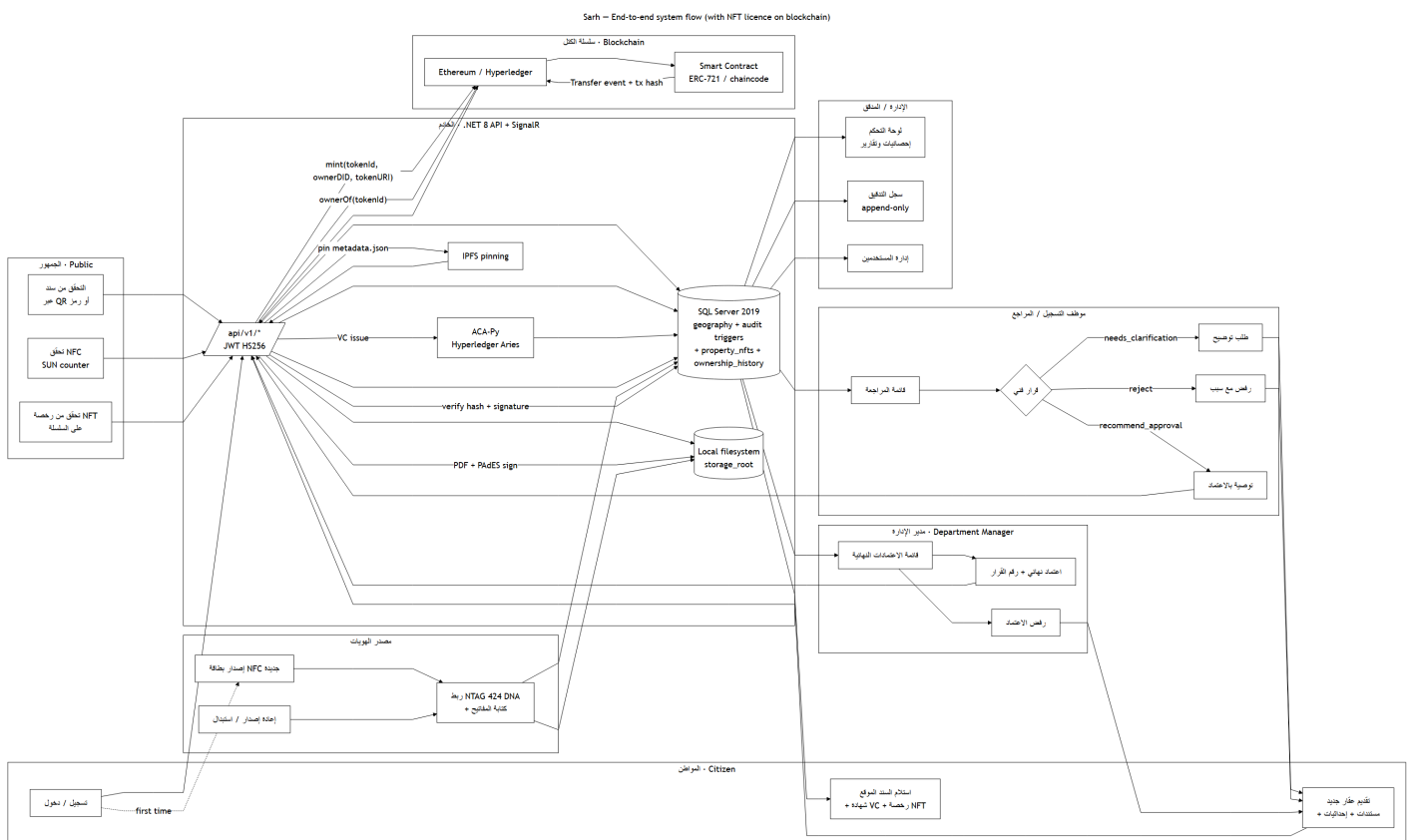
Sarh Platform as a single process with all external entities (Citizen, Registry officer, Department manager, SSI, Map, Notification, NFT minting service).



Five main processes (Account & Identity, Land Claim Processing, Spatial Auditing, NFT Documentation & Minting, Notifications) with data stores including D3 Blockchain Ledger.



Decomposition of process 3.0 — Document Review → Spatial Conflict Check → Technical Map Generation → Status Update.



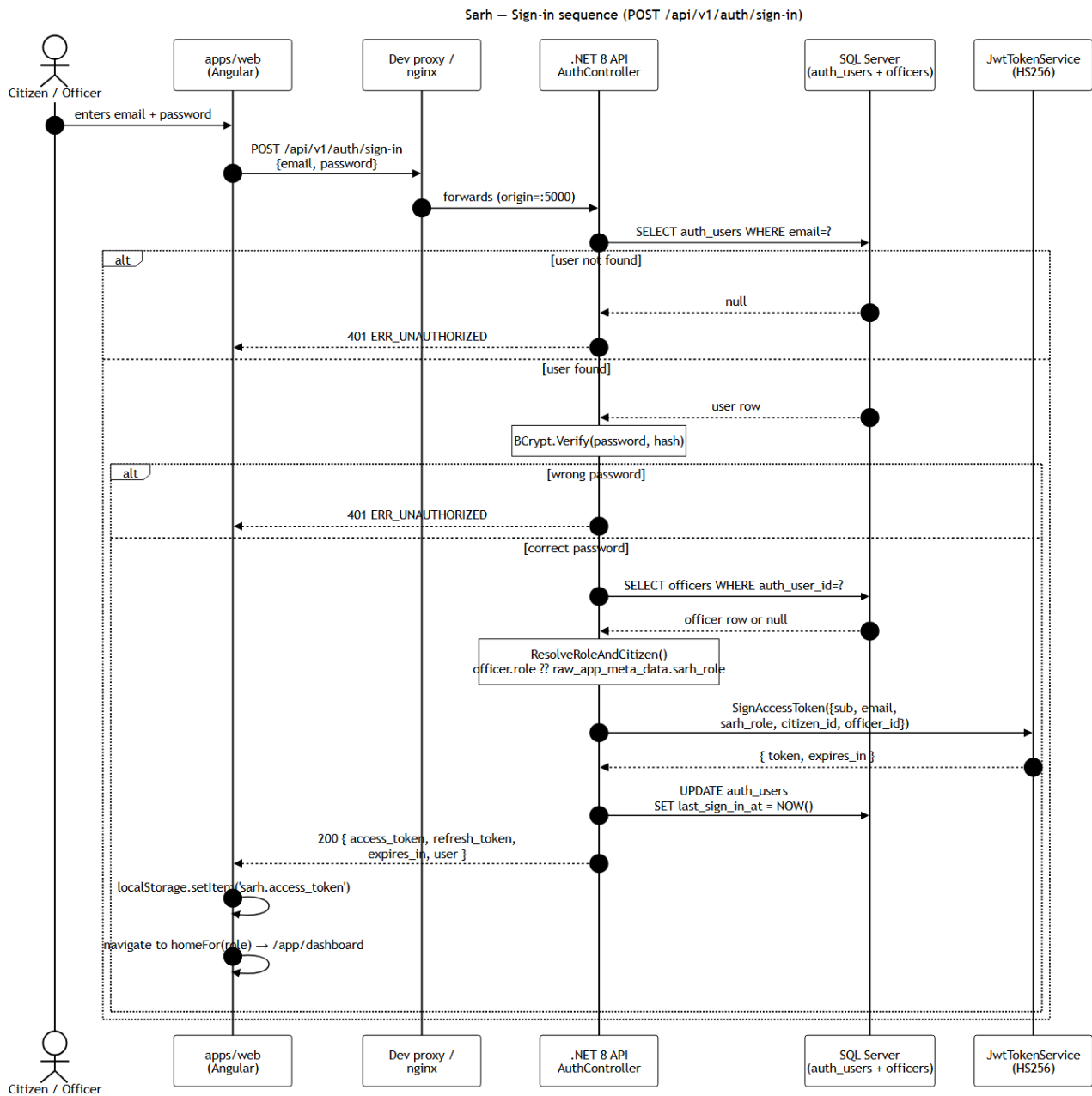
citizen → officer → manager → ID issuer → admin, with the .NET 8 API, SQL Server, ACA-Py, IPFS, and the blockchain at the centre.

05

مخططات التسلسل

SEQUENCE DIAGRAMS

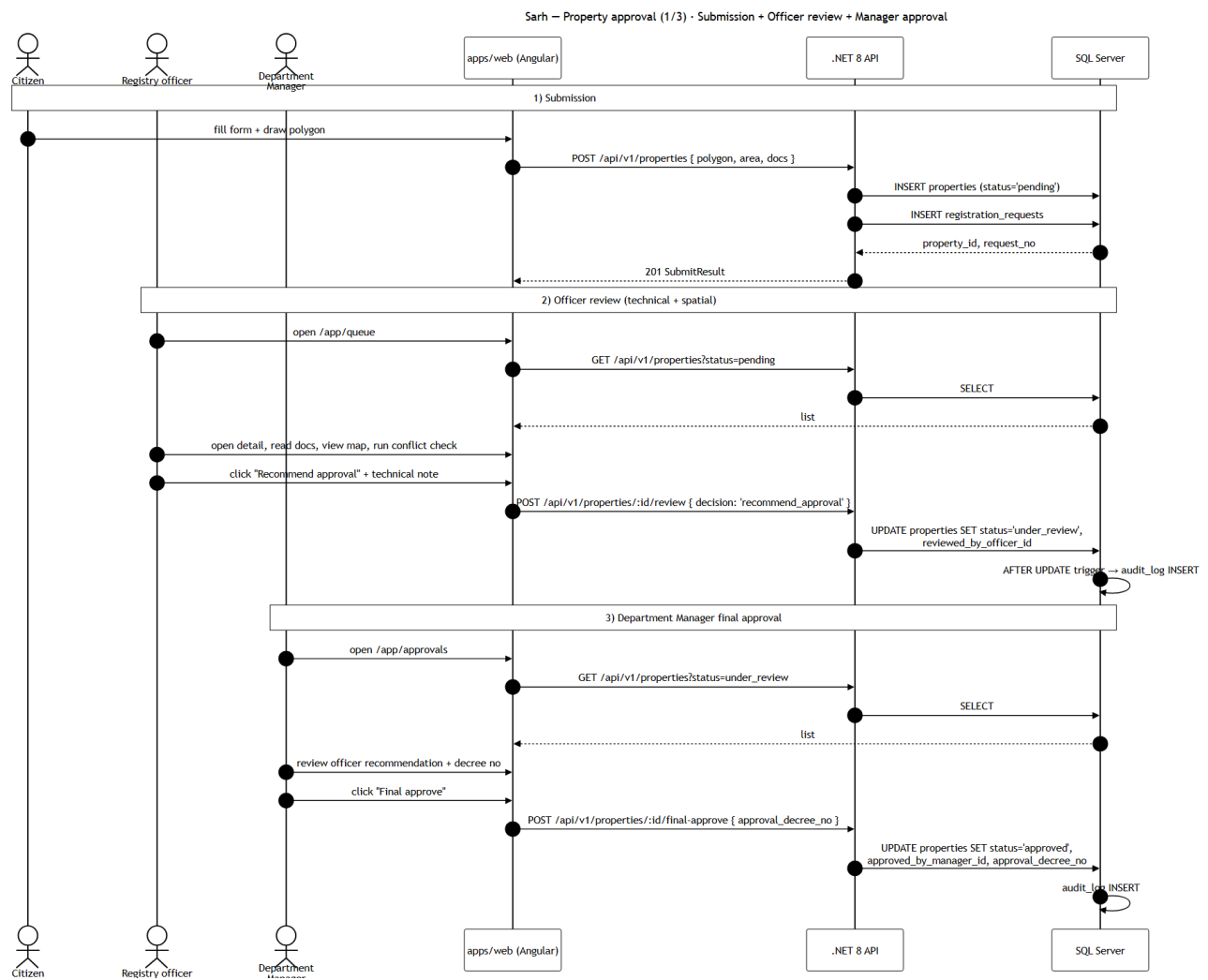
Step-by-step walk-throughs for sign-in, property approval (split into submit
./ mint / verify so each fits on A4), and ID-card issuance



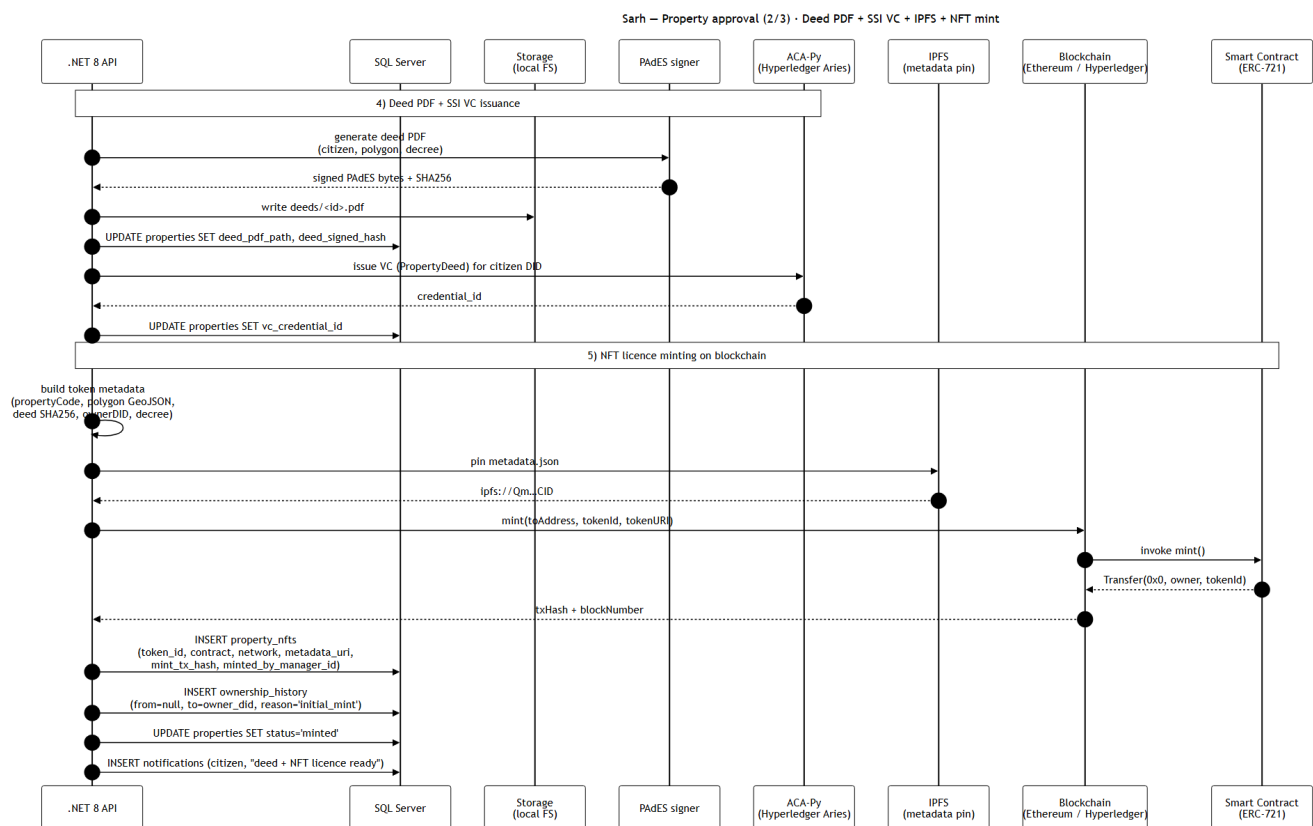
POST /api/v1/auth/sign-in from browser → proxy → AuthController → BCrypt.Verify → JWT issuance.

Property approval (1/3) · Submit + Review + Manager approval

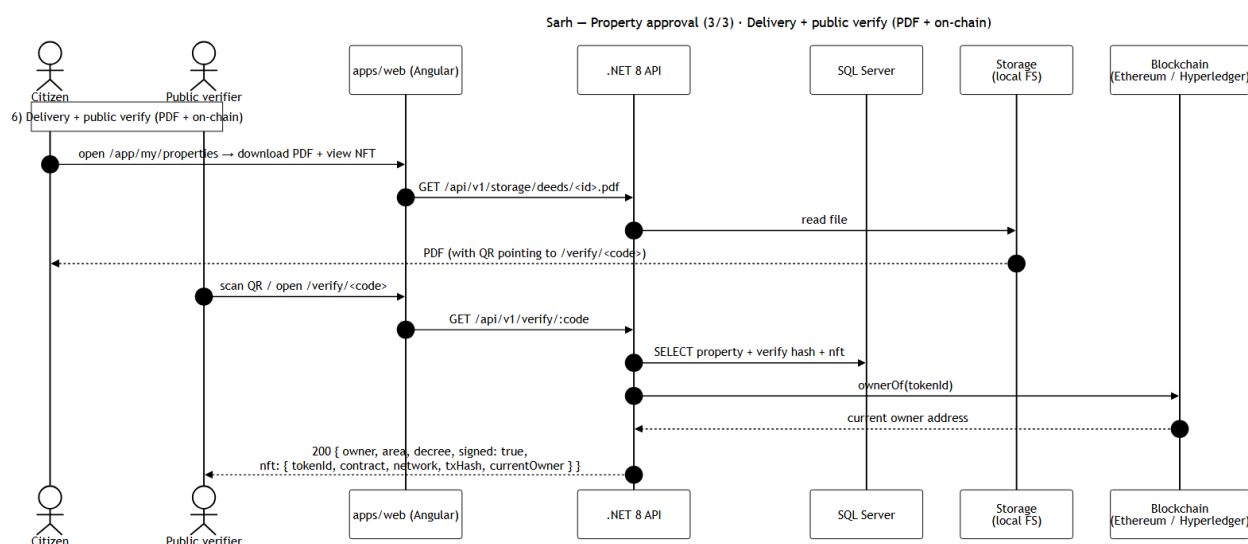
5.2 · موافقة العقار (١ / ٣) · التقديم والمراجعة والاعتماد



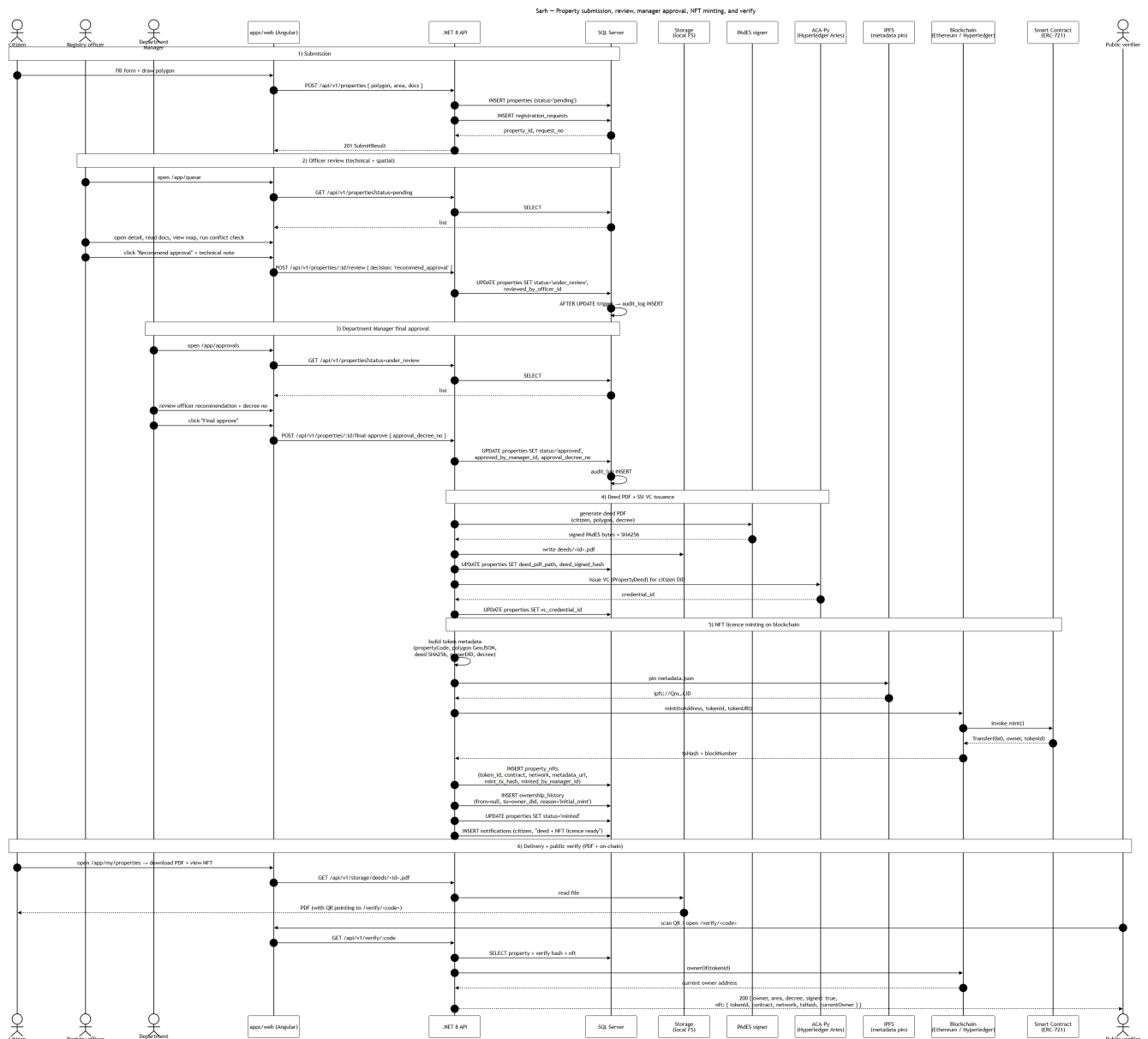
Citizen submission → Registry-officer recommendation → Department-manager final approval (with decree number).



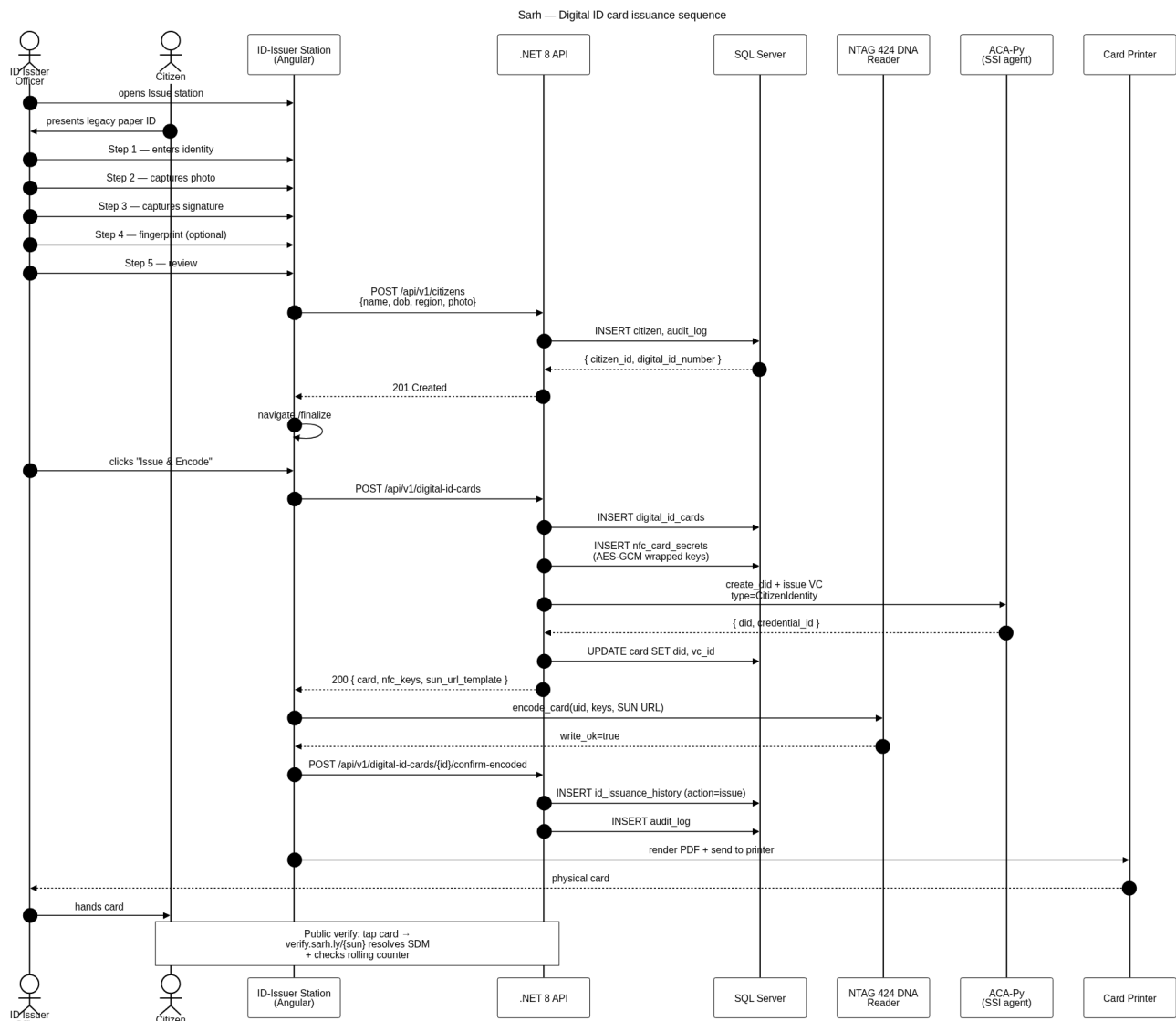
PAdES-signed deed → SSI VC issuance → IPFS pin → ERC-721 mint on-chain.



Citizen download (signed PDF + QR) → public verifier scans → API checks PDF hash + on-chain ownerOf.



Combined sequence including all 12 participants and 6 phases. Dense — split versions (5.2 – 5.4) are easier to read.



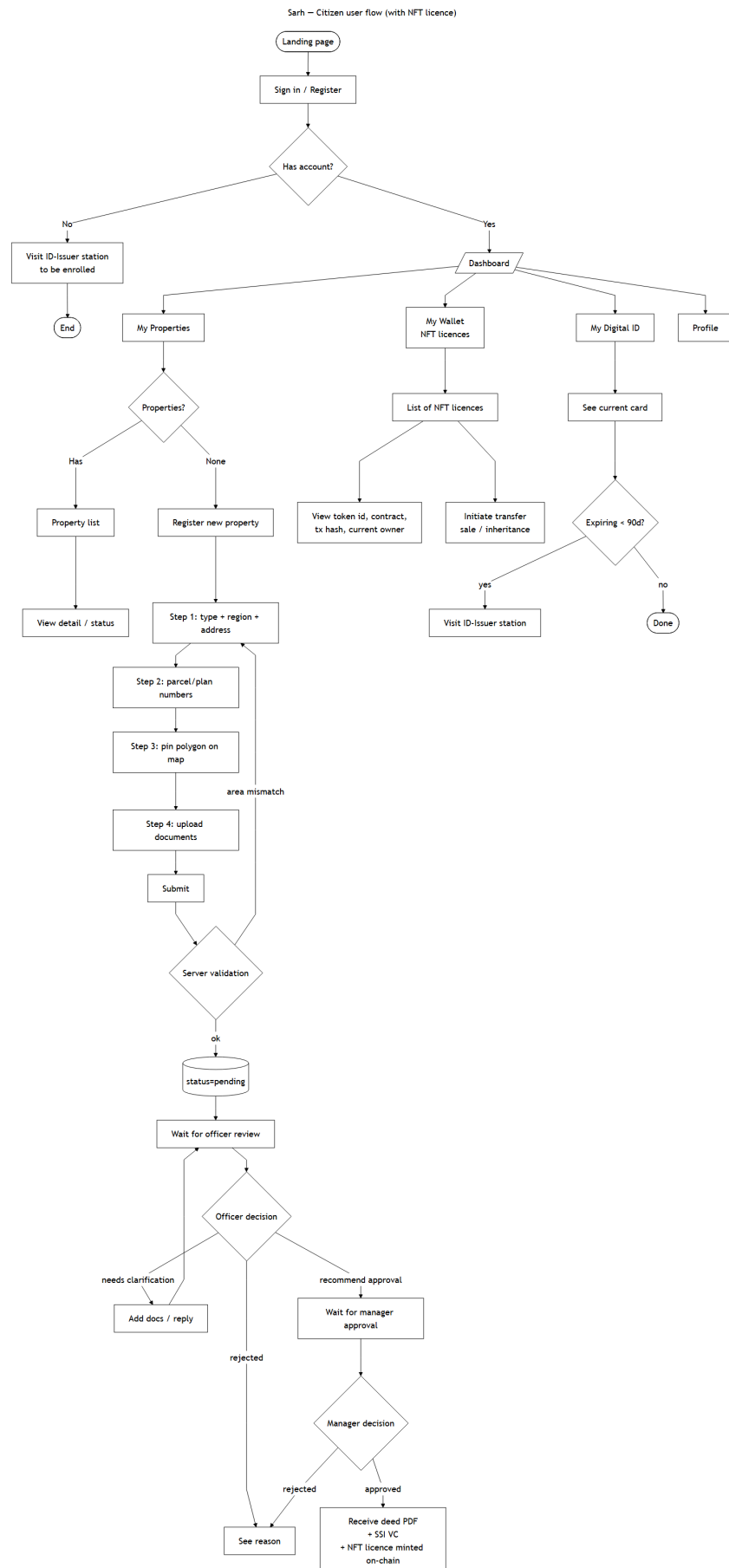
Wizard → citizen create → card issue → NFC encode → SSI VC → print.

06

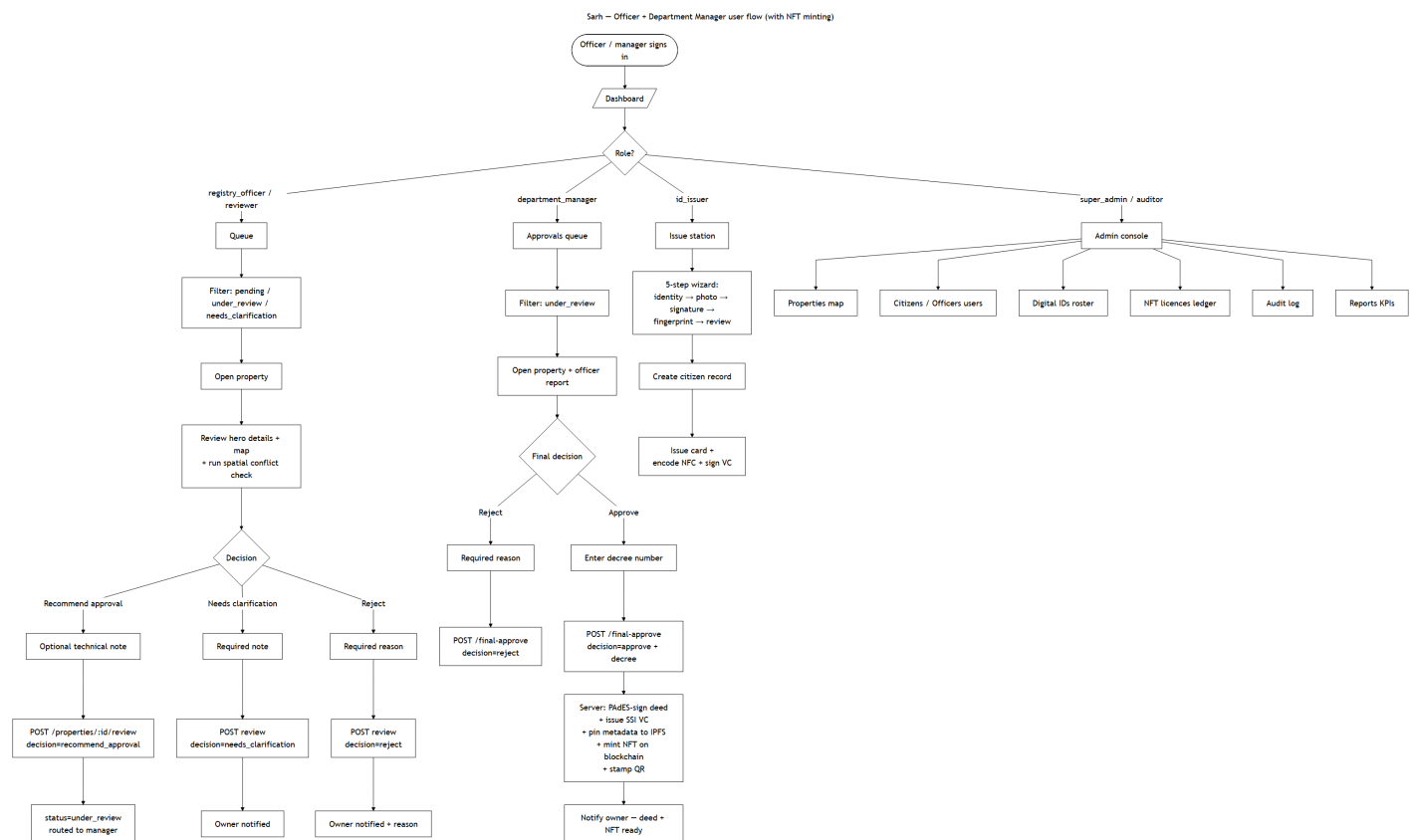
رحلات المستخدم

USER JOURNEYS

The two end-to-end journeys built into the platform — citizen registering a property and receiving an NFT licence, and officer/manager approving submissions and triggering minting



Landing → sign in → register property → wait for officer + manager → receive deed + NFT licence. Includes "My Wallet (NFT licences)" branch.



Officer queue → review → recommend; manager approval queue → final approve → triggers PADES + SSI + IPFS + mint chain.